

Chapter 5 Route Control

Lab 5-1 Route Import and Control

Learning Objectives

The objectives of this lab are to learn and understand:

- How to import OSPF and ISIS routes to each other
- How to configure route filtering based on IP prefix list
- How to filter routes using a route-policy

Topology

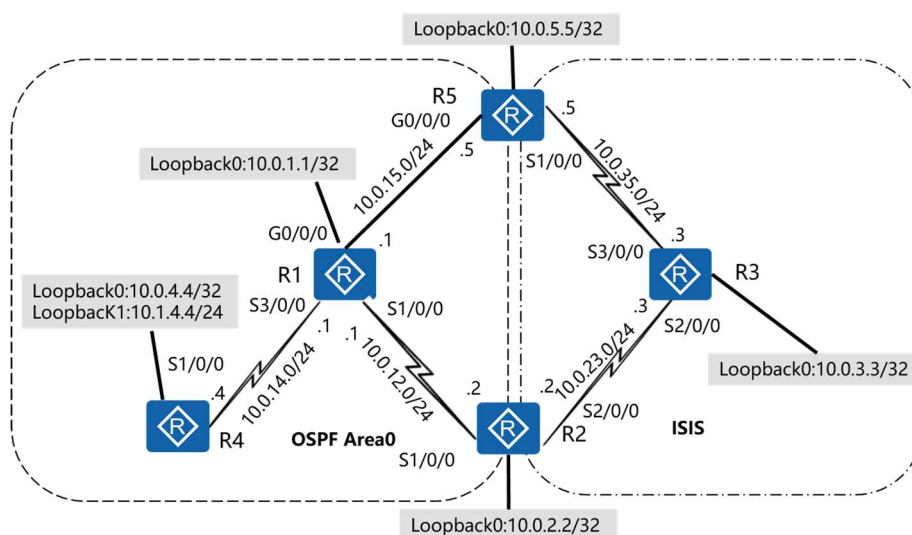


Figure 5-1 Route import and control

Scenario

You are a network administrator of a company. The company's network has OSPF areas and ISIS areas. To implement network connectivity, configure route import. When two routing protocols on two devices import routes from each other, some problems may occur, such as routing loops and sub-optimal routes. To prevent these problems, you can configure the IP prefix list and route-policy to control routes.

Tasks

Step 1 **Set basic parameters and configure IP addresses.**

Configure IP addresses and masks for physical interfaces and loopback interfaces of all routers. Each Loopback0 uses the 32-bit mask.

```
<R1>system-view
```

Enter system view, return user view with Ctrl+Z.

```
[R1]interface Serial 1/0/0
```

```
[R1-Serial1/0/0]ip address 10.0.12.1 255.255.255.0
```

```
[R1-Serial1/0/0]interface GigabitEthernet 0/0/0
```

```
[R1-GigabitEthernet0/0/0]ip address 10.0.15.1 255.255.255.0
```

```
[R1-GigabitEthernet0/0/0]interface Serial 3/0/0
```

```
[R1-Serial3/0/0]ip address 10.0.14.1 255.255.255.0
```

```
[R1-Serial3/0/0]interface LoopBack 0
```

```
[R1-LoopBack0]ip address 10.0.1.1 255.255.255.255
```

```
<R2>system-view
```

Enter system view, return user view with Ctrl+Z.

```
[R2]interface Serial 1/0/0
```

```
[R2-Serial1/0/0]ip address 10.0.12.2 255.255.255.0
```

```
[R2-Serial1/0/0]interface Serial 2/0/0
```

```
[R2-Serial2/0/0]ip address 10.0.23.2 255.255.255.0
```

```
[R2-Serial2/0/0]interface LoopBack 0
```

```
[R2-LoopBack0]ip add 10.0.2.2 255.255.255.255
```

```
<R3>system-view
```

Enter system view, return user view with Ctrl+Z.

```
[R3]interface Serial 2/0/0
```

```
[R3-Serial2/0/0]ip address 10.0.23.3 255.255.255.0
```

```
[R3-Serial2/0/0]interface Serial 3/0/0
[R3-Serial3/0/0]ip address 10.0.35.3 255.255.255.0
[R3-Serial3/0/0]interface LoopBack 0
[R3-LoopBack0]ip address 10.0.3.3 255.255.255.255
```

<R4>system-view

Enter system view, return user view with Ctrl+Z.

```
[R4]interface Serial 1/0/0
[R4-Serial1/0/0]ip address 10.0.14.4 255.255.255.0
[R4-Serial1/0/0]interface LoopBack 0
[R4-LoopBack0]ip address 10.0.4.4 255.255.255.255
```

<R5>system-view

Enter system view, return user view with Ctrl+Z.

```
[R5]interface Serial 1/0/0
[R5-Serial1/0/0]ip address 10.0.35.5 255.255.255.0
[R5-Serial1/0/0]interface GigabitEthernet 0/0/0
[R5-GigabitEthernet0/0/0]ip address 10.0.15.5 255.255.255.0
[R5-GigabitEthernet0/0/0]interface LoopBack 0
[R5-LoopBack0]ip address 10.0.5.5 255.255.255.255
```

After the configurations are complete, test link connectivity.

```
[R1]ping -c 1 10.0.12.2
```

```
PING 10.0.12.2: 56 data bytes, press CTRL_C to break
```

```
Reply from 10.0.12.2: bytes=56 Sequence=1 ttl=255 time=38 ms
```

```
--- 10.0.12.2 ping statistics ---
```

```
1 packet(s) transmitted
```

1 packet(s) received

0.00% packet loss

round-trip min/avg/max = 38/38/38 ms

[R1]ping -c 1 10.0.15.5

PING 10.0.15.5: 56 data bytes, press CTRL_C to break

Reply from 10.0.15.5: bytes=56 Sequence=1 ttl=255 time=12 ms

--- 10.0.15.5 ping statistics ---

1 packet(s) transmitted

1 packet(s) received

0.00% packet loss

round-trip min/avg/max = 12/12/12 ms

[R1]ping -c 1 10.0.14.4

PING 10.0.14.4: 56 data bytes, press CTRL_C to break

Reply from 10.0.14.4: bytes=56 Sequence=1 ttl=255 time=33 ms

--- 10.0.14.4 ping statistics ---

1 packet(s) transmitted

1 packet(s) received

0.00% packet loss

round-trip min/avg/max = 33/33/33 ms

[R2]ping -c 1 10.0.23.3

PING 10.0.23.3: 56 data bytes, press CTRL_C to break

Reply from 10.0.23.3: bytes=56 Sequence=1 ttl=255 time=34 ms

--- 10.0.23.3 ping statistics ---

1 packet(s) transmitted

1 packet(s) received

0.00% packet loss

round-trip min/avg/max = 34/34/34 ms

[R3]ping -c 1 10.0.35.5

PING 10.0.35.5: 56 data bytes, press CTRL_C to break

Reply from 10.0.35.5: bytes=56 Sequence=1 ttl=255 time=39 ms

--- 10.0.35.5 ping statistics ---

1 packet(s) transmitted

1 packet(s) received

0.00% packet loss

round-trip min/avg/max = 39/39/39 ms

Step 2 **Configure an IGP protocol.**

Run OSPF on R1, R2, R4, and R5. All devices belong to area 0.

Run OSPF on the network segments connected to R1's S1/0/0, S3/0/0, G0/0/0, and Loopback0.

[R1]ospf 1

[R1-ospf-1]area 0

[R1-ospf-1-area-0.0.0.0]network 10.0.12.1 0.0.0.0

[R1-ospf-1-area-0.0.0.0]network 10.0.15.1 0.0.0.0

[R1-ospf-1-area-0.0.0.0]network 10.0.14.1 0.0.0.0

[R1-ospf-1-area-0.0.0.0]network 10.0.1.1 0.0.0.0

Run OSPF on the network segment connected to R2's S1/0/0.

```
[R2]ospf 1
[R2-ospf-1]area 0
[R2-ospf-1-area-0.0.0.0]network 10.0.12.2 0.0.0.0
```

Run OSPF on the network segment connected to R4's S1/0/0 and Loopback0.

```
[R4]ospf 1
[R4-ospf-1]area 0
[R4-ospf-1-area-0.0.0.0]network 10.0.14.4 0.0.0.0
[R4-ospf-1-area-0.0.0.0]network 10.0.4.4 0.0.0.0
```

Run OSPF on the network segment connected to R5's G0/0/0.

```
[R5]ospf 1
[R5-ospf-1]area 0
[R5-ospf-1-area-0.0.0.0]network 10.0.15.5 0.0.0.0
```

Check whether the routers can learn the routes from the network segments connected to Loopback0 of other routers.

```
[R1]display ip routing-table
```

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 17 Routes : 17

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.4.4/32	OSPF	10	1562	D	10.0.14.4	Serial3/0/0

10.0.12.0/24	Direct	0	0	D	10.0.12.1	Serial1/0/0
10.0.12.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.12.2/32	Direct	0	0	D	10.0.12.2	Serial1/0/0
10.0.12.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.14.0/24	Direct	0	0	D	10.0.14.1	Serial3/0/0
10.0.14.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.14.4/32	Direct	0	0	D	10.0.14.4	Serial3/0/0
10.0.14.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.15.0/24	Direct	0	0	D	10.0.15.1	GigabitEthernet0/0/0
10.0.15.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.15.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

[R2]display ip routing-table

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 17 Routes : 17

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.1/32	OSPF	10	1562	D	10.0.12.1	Serial1/0/0
10.0.2.2/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.4.4/32	OSPF	10	3124	D	10.0.12.1	Serial1/0/0
10.0.12.0/24	Direct	0	0	D	10.0.12.2	Serial1/0/0
10.0.12.1/32	Direct	0	0	D	10.0.12.1	Serial1/0/0

10.0.12.2/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.12.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.14.0/24	OSPF	10	3124	D	10.0.12.1	Serial1/0/0
10.0.15.0/24	OSPF	10	1563	D	10.0.12.1	Serial1/0/0
10.0.23.0/24	Direct	0	0	D	10.0.23.2	Serial2/0/0
10.0.23.2/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.23.3/32	Direct	0	0	D	10.0.23.3	Serial2/0/0
10.0.23.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

[R4]display ip routing-table

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 12 Routes : 12

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.1/32	OSPF	10	1562	D	10.0.14.1	Serial1/0/0
10.0.4.4/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.12.0/24	OSPF	10	3124	D	10.0.14.1	Serial1/0/0
10.0.14.0/24	Direct	0	0	D	10.0.14.4	Serial1/0/0
10.0.14.1/32	Direct	0	0	D	10.0.14.1	Serial1/0/0
10.0.14.4/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.14.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.15.0/24	OSPF	10	1563	D	10.0.14.1	Serial1/0/0

127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

[R5]display ip routing-table

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 16 Routes : 16

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.1/32	OSPF	10	1	D	10.0.15.1	GigabitEthernet0/0/0
10.0.4.4/32	OSPF	10	1563	D	10.0.15.1	GigabitEthernet0/0/0
10.0.5.5/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.12.0/24	OSPF	10	1563	D	10.0.15.1	GigabitEthernet0/0/0
10.0.14.0/24	OSPF	10	1563	D	10.0.15.1	GigabitEthernet0/0/0
10.0.15.0/24	Direct	0	0	D	10.0.15.5	GigabitEthernet0/0/0
10.0.15.5/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.15.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.35.0/24	Direct	0	0	D	10.0.35.5	Serial1/0/0
10.0.35.3/32	Direct	0	0	D	10.0.35.3	Serial1/0/0
10.0.35.5/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.35.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

Configure ISIS on R2, R3, and R5.

Run ISIS on the network segments connected to R2's S2/0/0 and Loopback0.

```
[R2]isis 1
[R2-isis-1]network-entity 49.0001.0000.0000.0002.00
[R2-isis-1]is-level level-2
[R2-isis-1]interface serial2/0/0
[R2-Serial2/0/0]isis enable 1
[R2]interface loopback0
[R2-LoopBack0]isis enable 1
```

Run ISIS on the network segments connected to R3's S2/0/0, S3/0/0, and Loopback0.

```
[R3]isis 1
[R3-isis-1]network-entity 49.0001.0000.0000.0003.00
[R3-isis-1]is-level level-2
[R3-isis-1]interface serial2/0/0
[R3-Serial2/0/0]isis enable 1
[R3-Serial2/0/0]interface serial3/0/0
[R3-Serial3/0/0]isis enable 1
[R3-Serial3/0/0]interface loopback0
[R3-LoopBack0]isis enable 1
```

Run ISIS on the network segments connected to R5's S1/0/0 and Loopback0.

```
[R5]isis 1
[R5-isis-1]network-entity 49.0001.0000.0000.0005.00
[R5-isis-1]is-level level-2
[R5-isis-1]interface serial1/0/0
[R5-Serial1/0/0]isis enable 1
```

```
[R5-Serial1/0/0]interface loopback 0
```

```
[R5-LoopBack0]isis enable 1
```

Check whether the routers learn Loopback0 addresses of other devices.

```
[R2]display ip routing-table
```

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 20		Routes : 20				
Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.1/32	OSPF	10	48	D	10.0.12.1	Serial1/0/0
10.0.2.2/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.3.3/32	ISIS-L2	15	10	D	10.0.23.3	Serial2/0/0
10.0.4.4/32	OSPF	10	96	D	10.0.12.1	Serial1/0/0
10.0.5.5/32	ISIS-L2	15	20	D	10.0.23.3	Serial2/0/0
10.0.12.0/24	Direct	0	0	D	10.0.12.2	Serial1/0/0
10.0.12.1/32	Direct	0	0	D	10.0.12.1	Serial1/0/0
10.0.12.2/32	Direct	0	0	D	127.0.0.1	Serial1/0/0
10.0.12.255/32	Direct	0	0	D	127.0.0.1	Serial1/0/0
10.0.14.0/24	OSPF	10	96	D	10.0.12.1	Serial1/0/0
10.0.15.0/24	OSPF	10	49	D	10.0.12.1	Serial1/0/0
10.0.23.0/24	Direct	0	0	D	10.0.23.2	Serial2/0/0
10.0.23.2/32	Direct	0	0	D	127.0.0.1	Serial2/0/0
10.0.23.3/32	Direct	0	0	D	10.0.23.3	Serial2/0/0
10.0.23.255/32	Direct	0	0	D	127.0.0.1	Serial2/0/0
10.0.35.0/24	ISIS-L2	15	20	D	10.0.23.3	Serial2/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

255.255.255.255/32 Direct 0 0 D 127.0.0.1 InLoopBack0

[R3]dis ip routing-table

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 15 Routes : 15

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.2.2/32	ISIS-L2	15	10	D	10.0.23.2	Serial2/0/0
10.0.3.3/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.5.5/32	ISIS-L2	15	10	D	10.0.35.5	Serial3/0/0
10.0.23.0/24	Direct	0	0	D	10.0.23.3	Serial2/0/0
10.0.23.2/32	Direct	0	0	D	10.0.23.2	Serial2/0/0
10.0.23.3/32	Direct	0	0	D	127.0.0.1	Serial2/0/0
10.0.23.255/32	Direct	0	0	D	127.0.0.1	Serial2/0/0
10.0.35.0/24	Direct	0	0	D	10.0.35.3	Serial3/0/0
10.0.35.3/32	Direct	0	0	D	127.0.0.1	Serial3/0/0
10.0.35.5/32	Direct	0	0	D	10.0.35.5	Serial3/0/0
10.0.35.255/32	Direct	0	0	D	127.0.0.1	Serial3/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

[R5]display ip routing-table

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 19

Routes : 19

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.1/32	OSPF	10	1	D	10.0.15.1	GigabitEthernet0/0/0
10.0.2.2/32	ISIS-L2	15	20	D	10.0.35.3	Serial1/0/0
10.0.3.3/32	ISIS-L2	15	10	D	10.0.35.3	Serial1/0/0
10.0.4.4/32	OSPF	10	49	D	10.0.15.1	GigabitEthernet0/0/0
10.0.5.5/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.12.0/24	OSPF	0	49	D	10.0.15.1	GigabitEthernet0/0/0
10.0.14.0/24	OSPF	10	49	D	10.0.15.1	GigabitEthernet0/0/0
10.0.15.0/24	Direct	0	0	D	10.0.15.5	GigabitEthernet0/0/0
10.0.15.5/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
10.0.15.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
10.0.23.0/24	ISIS-L2	15	20	D	10.0.35.3	Serial1/0/0
10.0.35.0/24	Direct	0	0	D	10.0.35.5	Serial1/0/0
10.0.35.3/32	Direct	0	0	D	10.0.35.3	Serial1/0/0
10.0.35.5/32	Direct	0	0	D	127.0.0.1	Serial1/0/0
10.0.35.255/32	Direct	0	0	D	127.0.0.1	Serial1/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

Step 3 Configure a prefix list to filter routes.

Create static routes 1.1.1.1/32, 1.1.1.0/24, 1.1.1.0/25, 1.1.0.0/16, and 1.0.0.0/8 pointing to NULL 0 on R1. Run the **import-route static** command to import routes to OSPF.

```
[R1]ip route-static 1.1.1.1 255.255.255.255 NULL 0
```

```
[R1]ip route-static 1.1.1.0 255.255.255.0 NULL 0
```

```
[R1]ip route-static 1.1.1.0 255.255.255.128 NULL 0
```

```
[R1]ip route-static 1.1.0.0 255.255.0.0 NULL 0
```

```
[R1]ip route-static 1.0.0.0 255.0.0.0 NULL 0
```

```
[R1]ospf 1
```

```
[R1-ospf-1]import-route static
```

Check whether R4 receives the static routes added to R1.

```
[R4]display ip routing-table
```

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 17 Routes : 17

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
1.0.0.0/8	O_ASE	150	1	D	10.0.14.1	Serial1/0/0
1.1.0.0/16	O_ASE	150	1	D	10.0.14.1	Serial1/0/0
1.1.1.0/24	O_ASE	150	1	D	10.0.14.1	Serial1/0/0
1.1.1.0/25	O_ASE	150	1	D	10.0.14.1	Serial1/0/0
1.1.1.1/32	O_ASE	150	1	D	10.0.14.1	Serial1/0/0
10.0.1.1/32	OSPF	10	1562	D	10.0.14.1	Serial1/0/0
10.0.4.4/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.12.0/24	OSPF	10	3124	D	10.0.14.1	Serial1/0/0
10.0.14.0/24	Direct	0	0	D	10.0.14.4	Serial1/0/0
10.0.14.1/32	Direct	0	0	D	10.0.14.1	Serial1/0/0
10.0.14.4/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.14.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

10.0.15.0/24	OSPF	10	1563	D	10.0.14.1	Serial1/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

Configure the prefix list **pref_r1** on R1 to match the route 1.1.1.0/24.

```
[R1]ip ip-prefix pref_r1 index 10 permit 1.1.1.0 24 greater-equal 24 less-equal 24
```

Create the routing policy **policy_r1** and invoke the prefix list **pref_r1** to control static route import on R1.

```
[R1]route-policy policy_r1 permit node 10
[R1-route-policy]if-match ip-prefix pref_r1
[R1-route-policy]ospf
[R1-ospf-1]import-route static route-policy policy_r1
```

Check the routing table on R4.

```
[R4]display ip routing-table
```

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 13 Routes : 13

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
1.1.1.0/24	O_ASE	150	1	D	10.0.14.1	Serial1/0/0
10.0.1.1/32	OSPF	10	1562	D	10.0.14.1	Serial1/0/0

10.0.4.4/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.12.0/24	OSPF	10	3124	D	10.0.14.1	Serial1/0/0
10.0.14.0/24	Direct	0	0	D	10.0.14.4	Serial1/0/0
10.0.14.1/32	Direct	0	0	D	10.0.14.1	Serial1/0/0
10.0.14.4/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.14.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.15.0/24	OSPF	10	1563	D	10.0.14.1	Serial1/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

Step 4 Use the route policy to filter routes, avoiding loops.

Create Loopback 1 with address 10.1.4.4/24 on R4 and run the **import-route direct** command to import routes to OSPF.

```
[R4]interface LoopBack 1
[R4-LoopBack1]ip address 10.1.4.4 255.255.255.0
[R4-LoopBack1]ospf 1
[R4-ospf-1]import-route direct
```

Import OSPF routes on R2 to ISIS, and import ISIS routes on R5 to OSPF.

```
[R2]isis 1
[R2-isis-1]import-route ospf

[R5]ospf
[R5-ospf-1]import-route isis
```

Test the connectivity between R1 and 10.1.4.4.

[R1]ping 10.1.4.4

PING 10.1.4.4: 56 data bytes, press CTRL_C to break

Request time out

Request time out

Request time out

Request time out

Request time out

--- 10.1.4.4 ping statistics ---

5 packet(s) transmitted

0 packet(s) received

100.00% packet loss

The connection is abnormal. Check the routing table on R1.

[R1]display ip routing-table

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 28 Routes : 28

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
1.0.0.0/8	Static	60	0	D	0.0.0.0	NULL0
1.1.0.0/16	Static	60	0	D	0.0.0.0	NULL0
1.1.1.0/24	Static	60	0	D	0.0.0.0	NULL0
1.1.1.0/25	Static	60	0	D	0.0.0.0	NULL0
1.1.1.1/32	Static	60	0	D	0.0.0.0	NULL0
10.0.1.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.2.2/32	O_ASE	150	1	D	10.0.15.5	GigabitEthernet0/0/0

10.0.3.3/32	O_ASE	150	1	D	10.0.15.5	GigabitEthernet0/0/0
10.0.4.4/32	OSPF	10	1562	D	10.0.14.4	Serial3/0/0
10.0.5.5/32	O_ASE	150	1	D	10.0.15.5	GigabitEthernet0/0/0
10.0.12.0/24	Direct	0	0	D	10.0.12.1	Serial1/0/0
10.0.12.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.12.2/32	Direct	0	0	D	10.0.12.2	Serial1/0/0
10.0.12.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.14.0/24	Direct	0	0	D	10.0.14.1	Serial3/0/0
10.0.14.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.14.4/32	Direct	0	0	D	10.0.14.4	Serial3/0/0
10.0.14.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.15.0/24	Direct	0	0	D	10.0.15.1	GigabitEthernet0/0/0
10.0.15.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.15.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.23.0/24	O_ASE	150	1	D	10.0.15.5	GigabitEthernet0/0/0
10.0.35.0/24	O_ASE	150	1	D	10.0.15.5	GigabitEthernet0/0/0
10.1.4.0/24	O_ASE	150	1	D	10.0.15.5	GigabitEthernet0/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

The next hop of the route 10.1.4.0/24 on R1 is R5.

Check the 10.1.4.0/24 route in the routing tables on R2, R3, and R5.

[R2]display ip routing-table 10.1.4.0

Route Flags: R - relay, D - download to fib

Routing Table : Public

Summary Count : 1

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.1.4.0/24	O_ASE	150	1	D	10.0.12.1	Serial1/0/0

[R3]display ip routing-table 10.1.4.0

Route Flags: R - relay, D - download to fib

Routing Table : Public

Summary Count : 1

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.1.4.0/24	ISIS-L2	15	74	D	10.0.23.2	Serial2/0/0

[R5]display ip routing-table 10.1.4.0

Route Flags: R - relay, D - download to fib

Routing Table : Public

Summary Count : 1

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.1.4.0/24	ISIS-L2	15	84	D	10.0.35.3	Serial1/0/0

Run the **tracert** command on R1 to check the path to 10.1.4.4.

[R1]tracert 10.1.4.4

traceroute to 10.1.4.4(10.1.4.4), max hops: 30 ,packet length: 40,press CTRL_C to break

```
1 10.0.15.5 61 ms 2 ms 2 ms
2 10.0.35.3 29 ms 28 ms 29 ms
3 10.0.23.2 31 ms 36 ms 36 ms
4 10.0.12.1 34 ms 36 ms 36 ms
5 10.0.15.5 34 ms 37 ms 37 ms
6 10.0.35.3 55 ms 59 ms 59 ms
```

```
7 10.0.23.2 60 ms 66 ms 66 ms
8 10.0.12.1 63 ms 66 ms 66 ms
9 10.0.15.5 65 ms 67 ms 67 ms
```

A loop is detected.

The reason is: After route import is configured, R5 can learn the 10.1.4.0/24 route in both the ISIS and OSPF domains.

The ISIS routes have higher priority than OSPF external routes, so R5 uses the routes learned from the ISIS domain.

R1 can learn this route from both R5 and R4. The two routes are OSPF external routes. Therefore, the cost values of the two routes are compared. R1 is connected to R5 through GE links, which is better than the serial link between R1 and R4. Therefore, R1 uses the routes learned from R5, causing the loop.

Apply the route policy **policy_r5** to R5 and add tag 100 to the route 10.1.4.0/24.

```
[R5]acl number 2001
[R5-acl-basic-2001]rule 0 permit source 10.1.4.0 0.0.0.255
[R5-acl-basic-2001]route-policy add_tag permit node 10
[R5-route-policy]if-match acl 2001
[R5-route-policy]apply tag 100
[R5-route-policy]route-policy add_tag permit node 20
[R5-route-policy]ospf
[R5-ospf-1]import-route rip route-policy add_tag
```

Check OSPF routing information on R1.

```
[R1]display ospf routing
```

```
OSPF Process 1 with Router ID 10.0.12.1
```

Routing Tables

Routing for Network

Destination	Cost	Type	NextHop	AdvRouter	Area
10.0.1.1/32	0	Stub	10.0.1.1	10.0.12.1	0.0.0.0
10.0.12.0/24	1562	Stub	10.0.12.1	10.0.12.1	0.0.0.0
10.0.14.0/24	1562	Stub	10.0.14.1	10.0.12.1	0.0.0.0
10.0.15.0/24	1	Transit	10.0.15.1	10.0.12.1	0.0.0.0
10.0.4.4/32	1562	Stub	10.0.14.4	10.0.14.4	0.0.0.0

Routing for ASEs

Destination	Cost	Type	Tag	NextHop	AdvRouter
1.1.1.0/24	1	Type2	1	10.0.15.5	10.0.35.5
10.0.2.2/32	1	Type2	1	10.0.15.5	10.0.35.5
10.0.3.3/32	1	Type2	1	10.0.15.5	10.0.35.5
10.0.5.5/32	1	Type2	1	10.0.15.5	10.0.35.5
10.0.14.1/32	1	Type2	1	10.0.15.5	10.0.35.5
10.0.23.0/24	1	Type2	1	10.0.15.5	10.0.35.5
10.0.35.0/24	1	Type2	1	10.0.15.5	10.0.35.5
10.1.4.0/24	1	Type2	100	10.0.15.5	10.0.35.5

Total Nets: 13

Intra Area: 5 Inter Area: 0 ASE: 8 NSSA: 0

R1 has the 10.1.4.0/24 route with tag 100, indicating that the route is obtained from R5.

To address the loop problem, filter out the 10.1.4.0/24 route when R5 imports ISIS routes to OSPF.

Configure the route policy **route_delete** on R5 to control ISIS route import to OSPF.

```
[R5]route-policy route_delete deny node 10
[R5-route-policy]if-match acl 2001
[R5-route-policy]route-policy route_delete permit node 20
[R5-route-policy]ospf 1
[R5-ospf-1]import-route rip route-policy route_delete
```

Check the routing table on R1.

```
[R1]display ip routing-table
```

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 28 Routes : 28

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
1.0.0.0/8	Static	60	0	D	0.0.0.0	NULL0
1.1.0.0/16	Static	60	0	D	0.0.0.0	NULL0
1.1.1.0/24	Static	60	0	D	0.0.0.0	NULL0
1.1.1.0/25	Static	60	0	D	0.0.0.0	NULL0
1.1.1.1/32	Static	60	0	D	0.0.0.0	NULL0
10.0.1.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.2.2/32	O_ASE	150	1	D	10.0.15.5	GigabitEthernet0/0/0
10.0.3.3/32	O_ASE	150	1	D	10.0.15.5	GigabitEthernet0/0/0
10.0.4.4/32	OSPF	10	1562	D	10.0.14.4	Serial3/0/0
10.0.5.5/32	O_ASE	150	1	D	10.0.15.5	GigabitEthernet0/0/0
10.0.12.0/24	Direct	0	0	D	10.0.12.1	Serial1/0/0
10.0.12.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0

10.0.12.2/32	Direct	0	0	D	10.0.12.2	Serial1/0/0
10.0.12.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.14.0/24	Direct	0	0	D	10.0.14.1	Serial3/0/0
10.0.14.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.14.4/32	Direct	0	0	D	10.0.14.4	Serial3/0/0
10.0.14.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.15.0/24	Direct	0	0	D	10.0.15.1	GigabitEthernet0/0/0
10.0.15.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.15.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.23.0/24	O_ASE	150	1	D	10.0.15.5	GigabitEthernet0/0/0
10.0.35.0/24	O_ASE	150	1	D	10.0.15.5	GigabitEthernet0/0/0
10.1.4.0/24	O_ASE	150	1	D	10.0.14.4	Serial3/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

The routing information is correct.

Test the connectivity between R1 and 10.1.4.4.

[R1]ping 10.1.4.4

PING 10.1.4.4: 56 data bytes, press CTRL_C to break

Reply from 10.1.4.4: bytes=56 Sequence=1 ttl=255 time=33 ms

Reply from 10.1.4.4: bytes=56 Sequence=2 ttl=255 time=29 ms

Reply from 10.1.4.4: bytes=56 Sequence=3 ttl=255 time=29 ms

Reply from 10.1.4.4: bytes=56 Sequence=4 ttl=255 time=29 ms

Reply from 10.1.4.4: bytes=56 Sequence=5 ttl=255 time=29 ms

--- 10.1.4.4 ping statistics ---

5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 29/29/33 ms

Test the path from R1 to 10.1.4.4.

```
[R1]tracert 10.1.4.4  
  
tracert to 10.1.4.4(10.1.4.4), max hops: 30 ,packet length: 40,press CTRL_C to break  
  
1 10.0.14.4 61 ms 29 ms 29 ms
```

The loop is removed. R4 is the next hop of the 10.1.4.0/24 route on R1.

Step 5 Use the route-policy to change route priorities, avoiding loops.

Check the IP routing table of R5. Observe the next hop of the route 10.1.4.0/24.

```
[R5]display ip routing-table  
  
Route Flags: R - relay, D - download to fib  
  
-----  
  
Routing Tables: Public  
  
Destinations : 22      Routes : 22  
  
Destination/Mask    Proto    Pre  Cost    Flags NextHop          Interface  
  
1.1.1.0/24         ISIS-L2   15   84      D    10.0.35.3    Serial1/0/0  
10.0.1.1/32        OSPF     10    1      D    10.0.15.1    GigabitEthernet0/0/0  
10.0.2.2/32        ISIS-L2   15   20      D    10.0.35.3    Serial1/0/0  
10.0.3.3/32        ISIS-L2   15   10      D    10.0.35.3    Serial1/0/0  
10.0.4.4/32        OSPF     10   49      D    10.0.15.1    GigabitEthernet0/0/0  
10.0.5.5/32        Direct    0     0      D    127.0.0.1    LoopBack0
```

10.0.12.0/24	OSPF	10	49	D	10.0.15.1	GigabitEthernet0/0/0
10.0.14.0/24	OSPF	10	49	D	10.0.15.1	GigabitEthernet0/0/0
10.0.14.1/32	ISIS-L2	15	84	D	10.0.35.3	Serial1/0/0
10.0.15.0/24	Direct	0	0	D	10.0.15.5	GigabitEthernet0/0/0
10.0.15.5/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
10.0.15.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
10.0.23.0/24	ISIS-L2	15	20	D	10.0.35.3	Serial1/0/0
10.0.35.0/24	Direct	0	0	D	10.0.35.5	Serial1/0/0
10.0.35.3/32	Direct	0	0	D	10.0.35.3	Serial1/0/0
10.0.35.5/32	Direct	0	0	D	127.0.0.1	Serial1/0/0
10.0.35.255/32	Direct	0	0	D	127.0.0.1	Serial1/0/0
10.1.4.0/24	ISIS-L2	15	84	D	10.0.35.3	Serial1/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

Test the path from R5 to 10.1.4.4.

[R5]tracert 10.1.4.4

traceroute to 10.1.4.4(10.1.4.4), max hops: 30 ,packet length: 40,press CTRL_C to break

```

1 10.0.35.3 62 ms  24 ms  24 ms
2 10.0.23.2 43 ms  44 ms  44 ms
3 10.0.12.1 33 ms  33 ms  33 ms
4 10.0.14.4 74 ms  55 ms  55 ms

```

Although route filtering can fix the loop problem, R5 still learns the 10.1.4.0/24 route from the ISIS domain. The sub-optimal route problem still exists.

To fix both the loop problem and sub-optimal route problem, R5 must learn the 10.1.4.0/24 route from the OSPF domain.

Delete the policy **route_delete** on R5.

```
[R5]undo route-policy route_delete
```

Configure the route-policy **route_pref** on R5. Change the priority of the 10.1.4.0/24 route to 180, which is smaller than the priority of OSPF external routes.

```
[R5]route-policy route_pref permit node 10
```

```
[R5-route-policy]if-match acl 2001
```

```
[R5-route-policy]apply preference 180
```

Use the route-policy **route_pref** to control the ISIS routes imported into OSPF.

```
[R5]isis
```

```
[R5-isis-1]preference route-policy route_pref
```

Check the IP routing tables of R5 and R1. Observe the next hops of the 10.1.4.0/24 routes.

```
[R5]display ip routing-table 10.1.4.0
```

Route Flags: R - relay, D - download to fib

```
-----
```

Routing Table : Public						
Summary Count : 1						
Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.1.4.0/24	O_ASE	150	1	D	10.0.15.1	GigabitEthernet0/0/0

```
[R1]display ip routing-table 10.1.4.0
```

Route Flags: R - relay, D - download to fib

Routing Table : Public

Summary Count : 1

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.1.4.0/24	O_ASE	150	1	D	10.0.14.4	Serial3/0/0

Test the connectivity between R1 and 10.1.4.4.

[R1]ping 10.1.4.4

PING 10.1.4.4: 56 data bytes, press CTRL_C to break

Reply from 10.1.4.4: bytes=56 Sequence=1 ttl=255 time=39 ms

Reply from 10.1.4.4: bytes=56 Sequence=2 ttl=255 time=35 ms

Reply from 10.1.4.4: bytes=56 Sequence=3 ttl=255 time=35 ms

Reply from 10.1.4.4: bytes=56 Sequence=4 ttl=255 time=35 ms

Reply from 10.1.4.4: bytes=56 Sequence=5 ttl=255 time=35 ms

--- 10.1.4.4 ping statistics ---

5 packet(s) transmitted

5 packet(s) received

0.00% packet loss

round-trip min/avg/max = 35/35/39 ms

Test the path from R1 to 10.1.4.4.

[R1]tracert 10.1.4.4

tracert to 10.1.4.4(10.1.4.4), max hops: 30 ,packet length: 40,press CTRL_C to break

1 10.0.14.4 61 ms 25 ms 25 ms

Test the path from R5 to 10.1.4.4.

[R5]tracert 10.1.4.4

tracert to 10.1.4.4(10.1.4.4), max hops: 30 ,packet length: 40,press CTRL_C to break

1 10.0.15.1 61 ms 2 ms 2 ms

2 10.0.14.4 41 ms 28 ms 27 ms

The loop problem is fixed.

R4 is the next hop of the route 10.1.4.0/24 on R1. R1 is the next hop of the route 10.1.4.0/24 on R5. The sub-optimal route problem is also fixed.

----End